

KISHAN RAJ

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Portfolio | LeetCode | CodeChef | GitHub | LinkedIn

Education

Army Institute of Technology, Pune

2023 – 2027

Bachelor of Engineering (B.E.), Information Technology

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management System, Operating Systems, Computer Networks
- **CGPA:** 8.54/10

Experience

Open Source — AWS CDK (aws/aws-cdk)

Contributor — **GitHub**

- Resolved AWS ECS CloudFormation synthesis bug in the AWS CDK framework by implementing automated container deployment triggers during underlying EC2 **Auto Scaling Group** (ASG) instance mutations.
- Engineered a deterministic hashing mechanism injected into the **Ec2Service** construct to force **CloudFormation** to register strategy changes, validated with dedicated unit tests in the **aws-ecs** test suite.

Accenture Innovation Challenge

Team Lead — *Algorithm & Framework Development*

- Led a 5-member team to architect an automated eco-packaging routing framework, earning a **Top 10%** finish among 100,000+ national competitors.
- Designed a custom **graph algorithm** and high-fidelity prototype **API**, optimizing transit paths to cut carbon waste while handling high-throughput mock traffic data.

EY Techathon 5.0

Team Lead — *LLM & AI Driven Sessions*

- Engineered an AI-driven career pathing algorithm with Node.js, boosting roadmap accuracy by **40%** via **LangChain** powered competency mapping and **dynamic prompt** optimization.
- Integrated secure **MongoDB** schemas to capture LLM session history and skill assessments, utilizing **JWT**-authorized APIs and Jest to ensure high data reliability.

Technical Skills

Languages: C++, Java, JavaScript, SQL, Bash

Frameworks: Spring Boot, Node.js, React.js, Express.js

Databases: MongoDB, MySQL

DevOps & Infra: Docker, Kafka, gRPC, Terraform (IaC), CI/CD

Tools: Git, GitHub, Postman

Technical Projects

Distributed Collaborative IDE

React, WebSockets, Node.js, Express.js, MongoDB, REST APIs — **GitHub**

- Engineered a real-time collaborative code editor leveraging **WebSockets**, supporting multi-user live editing and instant synchronization, reducing latency by **15ms** across devices.
- Built a scalable backend execution engine with multi-language support and **MongoDB** session management, optimizing collaboration and reducing build times by **20%**.

Patient Management Portal

Spring Boot, Microservices, gRPC, Kafka, Docker, AWS — **GitHub**

- Architected a scalable microservices ecosystem using **gRPC** and Apache **Kafka** for event-driven streaming, reducing inter-service communication latency by **35%**.
- Engineered a secure API Gateway with **JWT Auth** and **CloudFormation** (IaC) for automated AWS deployment, cutting provisioning time by **50%**.

Achievements

- **LeetCode:** 1610 rating, **Top 15%** globally - 300 DSA problems solved. Profile
- **Distributed Collaborative IDE:** 350+ users, supporting real-time multi-user sessions in production. Live
- **Open Source (aws-cdk):** Merged a fix to **aws/aws-cdk** resolving GitHub issue of deployment triggers. Github PR
- **Data Science Training:** Completed a rigorous, hands-on training program conducted by IIT Guwahati. Certificate
- **IQube:** Secured **AIR 107** in the National IQ Examination. Certificate